

Pediatric DKA



Section I: Scenario Demographics

Scenario Title:	DKA and Decreased LOC		
Date of Development:	(11/09/2015)		
Target Learning Group:	☐ Juniors (PGY 1 - 2)	☒ Seniors (PGY $\geq$ 3)	☐ All Groups

Section II: Scenario Developers

Scenario Developer(s):	Lindsey McMurray
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Section III: Curriculum Integration

Learning Goals & Objectives	
Educational Goal:	To prioritize the management of an acutely unwell child.
CRM Objectives:	1) As team leader, to effectively communicate with the treating team 2) To manage an anxious parent witnessing a resuscitation
Medical Objectives:	1) To recognize and treat DKA 2) To secure an airway and appropriately ventilate a severely acidotic patient 3) To effectively run a PEA arrest 4) To manage a patient in ROSC

Case Summary: Brief Summary of Case Progression and Major Events
An 8 year old girl who has been tired and "unwell" for several days presents to the ED with an acute decline in her mental status. She is confused and lethargic. It becomes quickly apparent that the child is in DKA and requires immediate treatment. Due to decreasing neurologic status and vomiting, she eventually requires an advanced airway. The challenge is to optimize the peri-intubation course and to appropriately ventilate to allow for compensation of her metabolic acidosis.

References
Marx, J. A., Hockberger, R. S., Walls, R. M., & Adams, J. (2013). <i>Rosen's emergency medicine: Concepts and clinical practice</i> . St. Louis: Mosby. Chapter 126: Diabetes Mellitus and Disorders of Glucose Homeostasis.
EMCrit (2009). Intubating the patient with severe metabolic acidosis. Accessed on September 11, 2015 at http://emcrit.org/podcasts/tube-severe-acidosis/

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Section IV: Scenario Script

A. Clinical Vignette: To Read Aloud at Beginning of Case			
You have been called to the resuscitation bay to assess an 8 year old girl who has been brought in by her mother for lethargy and confusion. She has been unwell for 3 days with excessive fatigue, a few episodes of vomiting, and mild abdominal pain.			
B. Scenario Cast & Realism			
Patient:	☒ Computerized Mannequin ☐ Mannequin ☐ Standardized Patient ☐ Hybrid ☐ Task Trainer	Realism: <i>Select most important dimension(s)</i>	☒ Conceptual ☐ Physical ☐ Emotional/Experiential ☐ Other: ☐ N/A
Confederates		Brief Description of Role	
Mother	Gives past medical history, and history of presenting complaint. Becomes agitated if not addressed.		
C. Required Monitors			
☒ EKG Leads/Wires	☐ Temperature Probe	☐ Central Venous Line	
☒ NIBP Cuff	☐ Defibrillator Pads	☐ Capnography	
☒ Pulse Oximeter	☐ Arterial Line	☐ Other:	
D. Required Equipment			
☒ Gloves	☒ Nasal Prongs	☐ Scalpel	
☒ Stethoscope	☒ Venturi Mask	☐ Tube Thoracostomy Kit	
☐ Defibrillator	☒ Non-Rebreather Mask	☐ Cricothyroidotomy Kit	
☒ IV Bags/Lines	☒ Bag Valve Mask	☐ Thoracotomy Kit	
☒ IV Push Medications	☒ Laryngoscope	☐ Central Line Kit	
☐ PO Tabs	☒ Video Assisted Laryngoscope	☐ Arterial Line Kit	
☐ Blood Products	☒ ET Tubes	☐ Other:	
☐ Intraosseous Set-up	☐ LMA	☐ Other:	
E. Moulage			
None required.			
F. Approximate Timing			
Set-Up:	5 min	Scenario:	10 min
		Debriefing:	15 min

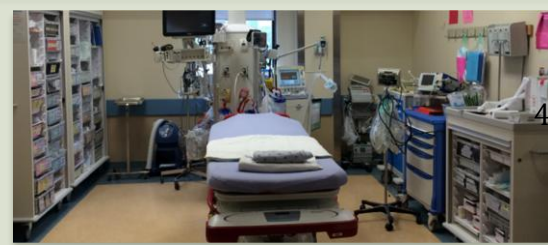
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Section V: Patient Data and Baseline State

A. Patient Profile and History					
Patient Name: Anna		Age: 8		Weight: 25kg	
Gender: ☐ M ☒ F		Code Status: Full			
Chief Complaint: Altered LOC					
History of Presenting Illness: Tired and "unwell" for 3 days with some mild abdominal pain and vomiting, now confused and lethargic.					
Past Medical History:	nil		Medications:	nil	
	Immunizations UTD				
Allergies: nil					
Social History: lives with mom and dad and 1 older brother.					
Family History: no family history.					
Review of Systems:	CNS:	Tired and lethargic.			
	HEENT:	Dry mucous membranes.			
	CVS:	Normal.			
	RESP:	Deep Kussmaul breathing.			
	GI:	Normal.			
	GU:	Polyuria.			
	MSK:	Normal.	INT:	Normal.	
B. Baseline Simulator State and Physical Exam					
☒ No Monitor Display		☐ Monitor On, no data displayed		☐ Monitor on Standard Display	
HR: 140/min	BP: 90/50	RR: 34/min (deep)	O ₂ SAT: 99%		
Rhythm: NSR	T: 36.5°C	Glucose: HIGH	GCS: 9 (E2 V2 M5)		
General Status: Appears unwell, altered mental status.					
CNS:	Moaning periodically, opens eyes to pain, withdraws to pain.				
HEENT:	Acetone breath, dry mucous membranes.				
CVS:	Tachycardic but normotensive.				
RESP:	High RR with deep Kusmaul breathing.				
ABDO:	Normal.				
GU:	Normal.				
MSK:	Normal.	SKIN:	Normal.		

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Section VI: Scenario Progression

Scenario States, Modifiers and Triggers			
Patient State	Patient Status	Learner Actions, Modifiers & Triggers to Move to Next State	
1. Baseline State Rhythm: Sinus tach HR: 140/min BP: 90/50 RR: 34/min O ₂ SAT: 99% T: 36.5°C	GCS 9-E2V2M5	<u>Learner Actions</u> - ☐ Monitor, full vitals - ☐ Check glucose (HIGH) - ☐ 2 large bore IVs - ☐ Take history from mom and reassure mom - ☐ Blood work (standard + VBG, lactate, serum ketones, osmols, lytes) - ☐ IV fluid bolus (10cc/kg) + maintenance (5-7ml/kg/hr) + KCl - ☐ Verbalize and prepare for possible intubation - ☐ IV insulin (0.1 U/kg/hr)	<u>Modifiers</u> <i>Changes to patient condition based on learner action</i> - Post fluid bolus, HR to 130, BP remains the same. - Nurse runs VBG immediately after it's ordered and notifies team leader of metabolic acidosis and normal K - If mom not addressed → mom to become more intrusive and difficult to have in room <u>Triggers</u> <i>For progression to next state</i> - 5 minutes → 2. Deterioration
2. Deterioration Rhythm: Sinus tach HR: 150/min BP: 60/30 RR: 34/min O ₂ SAT: 99% T: 36.5°C	GCS 7-E1V1M5 Vomiting and gagging on secretions.	<u>Learner Actions</u> - ☐ Recovery position, suction - ☐ Re-check glucose (24) - ☐ Decrease insulin infusion to 0.05 U/kg/hr - ☐ ± mannitol or hypertonic saline for possible cerebral edema - ☐ Elevate HOB - ☐ ± Pre-intubation bicarbonate - ☐ ± Pre-intubation 10cc/kg fluid bolus - ☐ Intubation - ☐ Push-dose vasopressor at bedside	<u>Modifiers</u> - No intubation by 2 min into state → O ₂ SAT slowly to 88% <u>Triggers</u> - Intubation with paralytic & resp rate not considered → 3. PEA Arrest - Intubation with resp rate considered → 4. ROSC
3. PEA Arrest Rhythm: PEA HR: no pulse BP: not detectable RR: BVM rate O ₂ SAT: no tracing T: 36.5°C	GCS 3	<u>Learner Actions</u> - ☐ Ensure quality CPR (15:2) - ☐ Epinephrine at 0.01mg/kg (.25mg) q 3-5 minutes. - ☐ Sodium bicarbonate (1meq/kg/dose ~ ½ amp) - ☐ Confirm tube placement - ☐ Go through Hs, Ts	<u>Modifiers</u> - If needed, after 1 cycle, RT will suggest ventilating at a faster rate to match pre-intubation RR <u>Triggers</u> - One CPR cycle after RR increased → 4. ROSC
4. ROSC Rhythm: Sinus tach HR: 140/min BP: 90/50 RR: 34/min (BVM) O ₂ SAT: 99% T: 36.5°C	GCS 3T	<u>Learner Actions</u> - ☐ Call PICU - ☐ Re-check electrolytes, gas - ☐ Post-intubation CXR - ☐ Insert NG/OG - ☐ Initiate sedation - ☐ ± Cooling (if had arrest)	<u>END CASE WITH PICU ARRIVAL</u>





Section VII: Supporting Documents, Laboratory Results, & Multimedia

Laboratory Results						
Na: 128	K: 4.5	Cl: 100	HCO ₃ : 8	BUN: 34	Cr: 60	Glu: 31
Ca: n/a		Mg: n/a		PO ₄ : n/a		Albumin: n/a
VBG	pH: 6.9	PCO ₂ : 20	PO ₂ : 50	HCO ₃ : 8	Lactate: 5	
WBC: 16		Hg: 127		Hct: 0.400		Plt: 400

Images (ECGs, CXRs, etc.)
CXR post intubation
Source: https://emcow.files.wordpress.com/2012/11/normal-intubation2.jpg



Section VIII: Debriefing Guide

General Debriefing Plan	
☐ Individual	☒ Group
☐ With Video	☒ Without Video
Objectives	
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Sample Questions for Debriefing	
1) How do you feel that your team communicated throughout this case? 2) How did it feel to have the mother in the room? Do you think the team handled this well? What are some approaches for having parents at the bedside during a resuscitation? 3) What is the differential for a child with altered LOC? 4) What is your initial management of DKA in a child, how does it differ from an adult? 5) What special considerations must you keep in mind when intubating a DKA patient or any patient with severe acidosis of any etiology? 6) If you suspect a PEA arrest secondary to acidosis, what are your treatment options?	
Key Moments	
Initiating management of DKA (primarily with fluids and then insulin).	
Recognizing the need to intubate, and vocalizing special considerations given severe acidosis.	
Running a PEA arrest.	